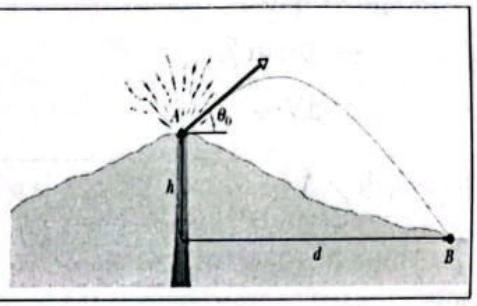
BRAC UNIVERSITY
Inspiring Excellence

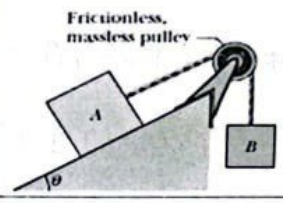
**MNS Department**
**Semester: Spring-2024; Section - 22**
**Course ID: PHY 111; Course Title: Principles of Physics I; QUIZ – 2; Total Marks: 15**

| Name: | ID: |
|---|---|

1. [7 marks] During volcanic eruptions, chunks of solid rock can be blasted out of the volcano; these projectiles are called *volcanic bombs*. Figure shows a cross section of Mt. Fuji, in Japan. A bomb is ejected at angle 42° to the horizontal, from the vent at *A* and dropped at the foot of the volcano at *B*, at vertical distance $h = 2.70\ km$ and horizontal distance $d = 8.30\ km$? Find the initial speed at A and the velocity of the bomb at B.




2. [5 marks] In Figure, Two blocks of mass $m_B = 2\ kg$ and $m_A = 7\ kg$ are connected by a massless string that passes over a frictionless pulley. Incline surface is frictionless and $\theta = 40.0°$. Draw force free body diagram, calculate the acceleration of the blocks with direction and the force on the block B from the surface.




3. [1 mark] A projectile is thrown from the ground. At where (position) velocity and acceleration perpendicular?

At maximum height ($H_{max}$)

4. [1 mark] At a certain instant, a fly ball has velocity $\vec{v} = 25\hat{\imath} - 8.6\hat{\jmath}$. Has the ball passed its highest point?

yes

5. [1 mark] In Fig. 2, what will be the acceleration of block B if you cut the string?

$-g$ or $-9.8\ ms^{-2}$

(1) We know, $x - x_0 = (V_0\cos\theta)t$

$$\Rightarrow t = \frac{x - x_0}{V_0\cos\theta} \quad \text{——— (1)}$$

$y_0 = 2.7Km = 2.7\times10^3 m$
$y = 0$
$x_0 = 0$
$x = d = 8.3Km = 8.3\times10^3 m$
$\theta = 42°$

Again, $y - y_0 = (V_0\sin\theta)t - \frac{1}{2}gt^2$

$$\Rightarrow y - y_0 = (V_0\sin\theta)\times\frac{x - x_0}{V_0\cos\theta} - \frac{g}{2}\times\frac{(x-x_0)^2}{(V_0\cos\theta)^2}$$

$$\Rightarrow 0 - 2.7\times10^3 = 8.3\times10^3\times\tan42° - \frac{9.8}{2}\times\frac{(8.3\times10^3)^2}{V_0^2(\cos42)^2}$$

$$\Rightarrow -2.7\times10^3 = 7.47\times10^3 - \frac{6.11\times10^8}{V_0^2}$$

$$\therefore V_0 = \boxed{245.16\ ms^{-1}}$$

$$(1) \Rightarrow t = \frac{8.3\times10^3}{245.16\times\cos42} = 45.56\ sec$$

$$\vec{V} = V_0\cos\theta\,\hat{\imath} + (V_0\sin\theta - gt)\,\hat{\jmath}$$
$$= 245.16\cos42°\,\hat{\imath} + (245.16\sin42° - 9.8\times45.56)\,\hat{\jmath}$$
$$\therefore \vec{V} = \boxed{182.19\,\hat{\imath} - 282.44\,\hat{\jmath}}$$

(2.) Here, $W_A \sin\theta = m_A g \sin\theta$

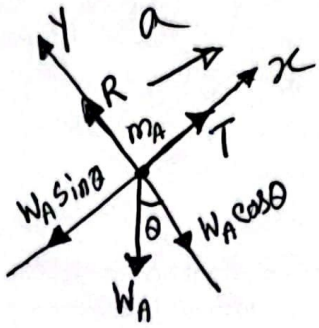
$= 7 \times 9.8 \times \sin 40^\circ$

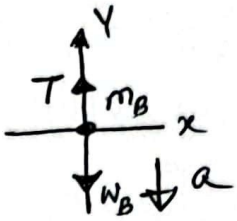
$= 44.1$

and, $W_B = m_B g$

$= 2 \times 9.8$

$= 19.6$




Since $W_A \sin\theta > W_B$, $\boxed{m_B \text{ is moving downward}}$

We Know, $F_{net,x} = m a_x$

For, $m_A \rightarrow T - W_A \sin\theta = m_A a$

$\Rightarrow T = 44.1 + 7a$ —— (1)

and for $m_B \rightarrow 0 = 0$

and, $F_{net,y} = m a_y$

For, $m_A \rightarrow R - W_A \cos\theta = m_A \times 0$

$\therefore R = W_A \cos\theta$

$= 52.55 N$

and for $m_B \rightarrow T - W_B = -m_B a$

$\Rightarrow 44.1 + 7a - 19.6 = -2a$ [From eqn (1)]

$\Rightarrow a = \frac{44.1 - 19.6}{9}$

$\therefore a = \boxed{2.72 ms^{-2}}$

$\boxed{\text{Force on B from the floor is zero}}$

Directions is worng in the answer.
mB will move upward